OPEN SOURCE AUDIT PROJECT

Analysis of Linux Kernel

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Course:
Open Source Software

A. Introduction:

Open source software plays a very important role in today's digital world. Many of the systems and applications we use daily are built on open source technologies. Unlike proprietary software, open source software allows users to view, modify, and distribute the source code freely.

This project focuses on understanding open source software by studying the Linux Kernel. The Linux Kernel is the core component of the Linux operating system and is widely used in servers, mobile devices, and embedded systems.

The aim of this project is to explore the origin, philosophy, licensing, and real-world impact of the Linux Kernel. Along with theoretical understanding, practical knowledge is demonstrated through shell scripts that perform basic Linux operations.

A1. Origin Story

The Linux Kernel was created by Linus Torvalds in 1991 when he was a student at the University of Helsinki. At that time, operating systems like UNIX were expensive and not freely available to students. Linus wanted a system that he could use and modify without restrictions.

He started developing a small kernel as a personal project and shared it on the internet. Unlike proprietary software, he allowed others to view and improve his code. This attracted developers from around the world, and gradually the project grew into a powerful and stable operating system kernel.

The idea of sharing code openly helped Linux gain popularity very quickly. It became a symbol of collaboration, where thousands of developers contributed to making it better. Today, the Linux Kernel is used in servers, Android devices, supercomputers, and many other systems.

A2. License(GPL v2)

The Linux Kernel is released under the GNU General Public License version 2 (GPL v2). This license ensures that the software remains free and open for everyone.

There are four important freedoms in free software:

1. The freedom to run the program for any purpose
2. The freedom to study how the program works
3. The freedom to modify the program
4. The freedom to distribute copies

The GPL license requires that if someone modifies the software and distributes it, they must also share their changes under the same license. This ensures that the software remains open and cannot be made completely proprietary.

In contrast, permissive licenses like MIT allow modifications to be made without requiring the source code to be shared. Personally, I believe GPL is better for community-driven projects because it protects openness.

A3. Ethics

Open source software raises important ethical questions about how software should be developed and shared. On one hand, making software open allows everyone to learn, improve, and innovate. It promotes collaboration and reduces dependency on large corporations.

However, there is also a debate about whether companies should be allowed to make profit using open source software. Many companies use Linux in their products and services while earning revenue from it. Some people believe this is fair because they add value, while others feel it takes advantage of community contributions.

The idea of “standing on the shoulders of giants” means building new innovations using existing knowledge. Open source strongly supports this idea by allowing developers to learn from existing code and improve upon it.

In my opinion, open source encourages innovation rather than limiting it. It creates an environment where knowledge is shared and improved continuously.

B. Linux Footprint:

The Linux Kernel is the core part of the operating system and interacts directly with the hardware. It does not exist as a single user application but works in the background to manage system resources.

In a Linux system, important directories include:

1. /etc: stores configuration files
2. /usr: contains user programs and libraries
3. /var/log: stores log files
4. /boot: contains kernel-related files

The kernel itself is loaded during system startup and runs with high privileges. It manages processes, memory, and devices.

The Linux system uses users and groups to maintain security. Most system processes run under restricted users to prevent unauthorized access.

Linux services can be managed using commands like start, stop, and restart. Updates to the kernel are provided by the open-source community and are regularly improved for performance and security.

C. FOSS ecosystem:

The Linux Kernel does not work alone. It is part of a larger open-source ecosystem that includes many tools and utilities.

It depends on components such as compilers (like GCC), libraries, and system tools that are part of the GNU project. Together, they form a complete operating system.

The Linux ecosystem has enabled the development of many technologies such as web servers, databases, and cloud platforms. It is widely used in the LAMP stack (Linux, Apache, MySQL, PHP), which powers many websites.

There is a large global community behind Linux, including developers, companies, and organizations. Contributions are made through mailing lists, forums, and version control platforms. This collaborative environment helps in continuous improvement.

D. Comparison:

Dimension	Linux (Open Source)	Windows (Proprietary)
Cost	Free to use	Paid license required
Security	Open code, can be audited	Closed source
Support	Community support	Official company support
Modification	Fully customizable	Limited customization
Control	User has full control	Controlled by company

Conclusion:

After comparing open source and proprietary software, it is clear that both have their advantages. Linux provides freedom, transparency, and flexibility, while Windows offers ease of use and official support.

In my opinion, Linux is more suitable for learning, development, and server environments because of its open nature. I would recommend using Linux for situations where customization and control are important.

I would also be interested in contributing to open source projects in the future, as it helps in learning and sharing knowledge with others.

References:

1. Official Linux Kernel documentation

2. GNU Project – Free Software Definition

3. Open Source Initiative (OSI)

4. Online articles and tutorials on Linux and open source software